



College of —
Computing and
Mathematical Sciences

From Quasi-Normal Modes to a Convergence Theorem for Chebyshev Collocation

A real research example in spectral methods

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Roadmap

We do **not** prove in one step that a spectral method works. We build a chain:

continuous QNM problem $\longrightarrow$ continuous spectral theory $\longrightarrow$ Chebyshev collocation
 $\longrightarrow$ consistency $\longrightarrow$ contour comparison $\longrightarrow$ local spectral convergence

What You Should See Today:

1. How the continuous problem is formulated
2. Which continuous properties are needed
3. How the collocation method is defined
4. What consistency means
5. How consistency leads to a convergence theorem

Why QNMs Are Not a Standard Eigenvalue Problem

Standard eigenvalue problem:

$$Au = \lambda u$$

- λ real
- Self-adjoint structure
- Orthogonality
- Variational tools

QNM problem:

$$\mathcal{P}(\omega)u = 0, \quad \omega \in \mathbb{C}$$

with

$$\mathcal{P}(\omega) = \mathcal{L}_0 + i\omega\mathcal{L}_1 + \omega^2\mathcal{L}_2$$

- ω generally **complex**
- **Quadratic** dependence on ω
- **Non-self-adjoint**
- Left $\neq$ right eigenfunctions

Consequence: The correct framework is the theory of **operator pencils**, not self-adjoint linear operators.

Continuous Problem Statement

Let $D \hookrightarrow X$ be Banach spaces, with X a Hilbert space.

Consider the **quadratic operator pencil**

$$\mathcal{P}(\omega) = \mathcal{L}_0 + i\omega\mathcal{L}_1 + \omega^2\mathcal{L}_2, \quad \omega \in \Lambda \subset \mathbb{C},$$

where $\mathcal{L}_j : D \rightarrow X$, $j = 0, 1, 2$.

We seek $(\omega, u) \in \mathbb{C} \times (D \setminus \{0\})$ such that $\mathcal{P}(\omega)u = 0$.

The physical boundary conditions are already encoded in the choice of the unknown and in the admissible space D .

Reduced problem: Find $(\omega, u) \in \mathbb{C} \times (D \setminus \{0\})$ such that $\mathcal{P}(\omega)u = 0$.

Kernel, Range, and Codimension

Let $A : V \rightarrow W$ be a linear operator.

Kernel (failure of injectivity):

$$\ker A = \{u \in V : Au = 0\}$$

Range (how much of W is reached):

$$\operatorname{ran} A = \{Au : u \in V\} \subset W$$

Codimension of the range (failure of surjectivity):

$$\operatorname{codim}(\operatorname{ran} A) := \dim(W / \operatorname{ran} A)$$

Counts how many independent directions in W are *missing* from the image of A .

Two basic defects: $\dim \ker A$ (injectivity) and $\operatorname{codim}(\operatorname{ran} A)$ (surjectivity)

Fredholm index: $\operatorname{ind}(A) = \dim \ker A - \operatorname{codim}(\operatorname{ran} A)$

Finite Dimensions: Rank-Nullity

Suppose $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$. Then rank-nullity gives:

$$n = \dim \ker A + \dim \operatorname{ran} A$$

Since $\operatorname{codim}(\operatorname{ran} A) = n - \dim \operatorname{ran} A$, we get $\operatorname{codim}(\operatorname{ran} A) = \dim \ker A$.

Therefore:

$$\operatorname{ind}(A) = 0 \quad \text{for **every** square matrix.}$$

In finite-dimensional linear algebra, the following are **equivalent**: A injective $\iff A$ surjective $\iff A$ invertible.

Students internalize this very strongly — and that is exactly why infinite-dimensional examples are so important pedagogically.

Infinite Dimensions: What Changes

In infinite dimensions, injectivity and surjectivity are **no longer tied together**.

Example 1: Injective but not surjective

Right shift on ℓ^2 :

$$S(x_1, x_2, x_3, \dots) = (0, x_1, x_2, \dots)$$

- $\ker S = \{0\}$ (injective)
- $\text{ran } S = \{y \in \ell^2 : y_1 = 0\}$
- $\text{codim}(\text{ran } S) = 1$
- $\text{ind}(S) = 0 - 1 = -1$

Example 2: Surjective but not injective

Left shift on ℓ^2 :

$$L(x_1, x_2, x_3, \dots) = (x_2, x_3, x_4, \dots)$$

- $\ker L = \{(x_1, 0, 0, \dots)\}$, $\dim \ker L = 1$
- L is surjective
- $\text{ind}(L) = 1 - 0 = 1$

“Injective $\iff$ surjective” is a **finite-dimensional phenomenon**, not a general one.

Why Closed Range Matters

There is another pathology that does not occur in finite dimensions: the range may **fail to be closed**.

Example: Injective, dense range, but not onto

On $L^2(0, 1)$, define $(Au)(x) = x \cdot u(x)$.

- A is injective: $xu(x) = 0$ a.e. implies $u = 0$ a.e.
- $\text{ran } A$ is **not closed**
- $\text{ran } A$ is dense but proper in $L^2(0, 1)$

“Almost onto” is **not** the same as onto. Dense range is not enough for good solvability theory.

This is why the definition of a Fredholm operator includes the requirement that the range be **closed**.

Range Versus Adjoint Kernel

In Hilbert spaces there is a fundamental relation:

$$(\operatorname{ran} A)^\perp = \ker A^*$$

If $\operatorname{ran} A$ is closed: $\operatorname{ran} A = (\ker A^*)^\perp$ and $\operatorname{codim}(\operatorname{ran} A) = \dim \ker A^*$

This means:

- **Right** null vectors ($\ker A$) measure failure of **injectivity**
- **Left** null vectors ($\ker A^*$) measure failure of **surjectivity**

For the QNM pencil $\mathcal{P}(\omega)$ at a fixed ω :

- $\ker \mathcal{P}(\omega)$ gives the **right** eigenfunctions
- $\ker \mathcal{P}(\omega)^*$ gives the **left** eigenfunctions
- $\operatorname{codim}(\operatorname{ran} \mathcal{P}(\omega)) = \dim \ker \mathcal{P}(\omega)^*$ when the range is closed

Left eigenfunctions are **not** an auxiliary construction. They encode the missing solvability conditions.

Summary: Why Fredholm Theory Appears

Finite-dimensional square matrix

$$\dim \ker A = \operatorname{codim}(\operatorname{ran} A)$$

So: injective $\iff$ surjective. Everything is automatic.

Infinite-dimensional operator

- Injective $\nRightarrow$ surjective
- Surjective $\nRightarrow$ injective
- $\operatorname{ran} A$ may fail to be closed

Nothing is automatic.

A **Fredholm operator** has finite-dimensional kernel and cokernel, and closed range — “almost invertible”.

If $\operatorname{ind}(A) = 0$: $\dim \ker A = \operatorname{codim}(\operatorname{ran} A)$, so injectivity $\iff$ surjectivity.

Fredholm of index 0 is the infinite-dimensional substitute for a square matrix.

Fredholm Operators, Index, and the Resolvent

Fredholm operator $A : D \rightarrow X$

- $\ker A$ is finite-dimensional
- $\operatorname{ran} A$ is closed
- $\operatorname{codim} \operatorname{ran} A$ is finite

Index: $\operatorname{ind}(A) = \dim \ker A - \operatorname{codim} \operatorname{ran} A$

If $\operatorname{ind}(A) = 0$: injectivity and surjectivity are closely linked.

Resolvent of a pencil

Resolvent set:

$$\rho(\mathcal{P}) = \{\omega \in \Lambda : \mathcal{P}(\omega) \text{ is invertible}\}$$

Spectrum:

$$\sigma(\mathcal{P}) = \Lambda \setminus \rho(\mathcal{P})$$

For $\omega \in \rho(\mathcal{P})$, the inverse $\mathcal{P}(\omega)^{-1} : X \rightarrow D$ is the **resolvent operator**.

Holomorphic resolvent: The map $\omega \mapsto \mathcal{P}(\omega)^{-1}$ is holomorphic as an operator-valued function on the open set $\rho(\mathcal{P})$.

- $\rho(\mathcal{P})$ is an open subset of $\mathbb{C}$; $\mathcal{P}(\omega)^{-1}$ is an operator for each $\omega \in \rho(\mathcal{P})$; the resolvent map is holomorphic on that set.

Adjoint Pencil, Left Eigenfunctions, Transversality

Right eigenfunction:

$$\mathcal{P}(\omega_0)u_0 = 0$$

Left eigenfunction:

$$\mathcal{P}(\omega_0)^*v_0 = 0$$

Because the problem is non-self-adjoint, we need **both** u_0 and v_0 .

Transversality quantity for a simple isolated eigenvalue ω_0 :

$$\tau_0 = (\mathcal{P}'(\omega_0)u_0, v_0)_X$$

If $\tau_0 \neq 0$: the eigenvalue is locally nondegenerate; τ_0 appears in perturbation formulas, condition numbers, and error estimates.

For non-self-adjoint problems, $(\mathcal{P}'(\omega_0)u_0, v_0)$ plays the role that orthogonality often plays in self-adjoint theory.

Adjoint pencil:

If $\mathcal{P}(\omega) = \mathcal{L}_0 + i\omega\mathcal{L}_1 + \omega^2\mathcal{L}_2$, then:

$$\mathcal{P}(\omega)^* = \mathcal{L}_0^* - i\bar{\omega}\mathcal{L}_1^* + \bar{\omega}^2\mathcal{L}_2^*$$

What Does “Transversality” Mean?

Scalar analogy:

If $f(\omega) = 0$ has a simple root at ω_0 , then $f'(\omega_0) \neq 0$.

So: $f'(\omega_0) \neq 0$ means a scalar root is simple; $\tau_0 \neq 0$ means a QNM eigenvalue is locally nondegenerate.

What does $\tau_0 \neq 0$ give us?

1. **No hidden Jordan chain at first order** — the eigenvalue is transverse, not tangential
2. **Perturbation formula:** $\delta\omega \approx \frac{(\delta\mathcal{P}(\omega_0)u_0, v_0)_X}{(\mathcal{P}'(\omega_0)u_0, v_0)_X}$
3. **Condition number:** $\kappa(\omega_0) = \frac{\|u_0\|_D \|v_0\|_X}{|\tau_0|}$
4. **Error estimates:** the denominator in the eigenvalue-error bound is controlled by $|\tau_0|$

$\tau_0 = (\mathcal{P}'(\omega_0)u_0, v_0)_X$ is the operator-pencil version of “nonzero derivative at a simple root”

Operator pencil:

The condition $\tau_0 \neq 0$ plays the same role, where

$$\tau_0 := (\mathcal{P}'(\omega_0)u_0, v_0)_X$$

What If Transversality Fails?

If $\tau_0 = 0$, the eigenvalue is no longer “simple in the right way”. This may indicate:

- Loss of semisimplicity
- Onset of a Jordan chain
- Branch collision
- Fold or exceptional-point behaviour under parameter variation

τ_0 is the quantity that links:

**Spectral
structure**

**Perturbation
theory**

Conditioning

**Error
bounds**

Main takeaway: Transversality $\tau_0 \neq 0$ is the fundamental nondegeneracy condition for operator pencils. When it fails, the local spectral structure becomes qualitatively different — Jordan chains, branch points, and loss of simple convergence behaviour.

Simple, Semisimple, Defective

Let ω_0 be an isolated eigenvalue of $\mathcal{P}(\omega)u = 0$.

Right eigenspace: $E_R(\omega_0) = \ker \mathcal{P}(\omega_0)$

Left eigenspace: $E_L(\omega_0) = \ker \mathcal{P}(\omega_0)^*$

Geometric multiplicity: $m_g(\omega_0) :=$

$\dim E_R(\omega_0)$

Algebraic multiplicity: $m_a(\omega_0) = \text{rank of}$
the local contour projector (accounts for
Jordan chains)

Simple eigenvalue:

$$m_a(\omega_0) = m_g(\omega_0) = 1$$

Semisimple eigenvalue:

$$m_a(\omega_0) = m_g(\omega_0) \geq 1$$

Several eigenfunctions possible, but **no**
Jordan chains.

Defective eigenvalue:

$$m_a(\omega_0) > m_g(\omega_0)$$

Nontrivial **Jordan chains** are present.

Simple = one eigenvalue, one eigenfunction, no hidden structure.

Semisimple = several eigenfunctions, but still no Jordan structure.

Defective = fewer eigenfunctions than algebraic multiplicity $\rightarrow$ generalized eigenvectors.

Why Simple vs Semisimple vs Defective Matters

Simple QNM

- One u_0 , one v_0
 - Transversality $\tau_0 \neq 0$
 - One nearby discrete eigenvalue $\omega_n \rightarrow \omega_0$
- Cleanest case.

Semisimple QNM

- Whole local eigenspace
- Correct object: **spectral projector** and its range
- Multiplicity preserved
- Local object is a **subspace**, not a single vector

Defective QNM

- Eigenspace too small
- Needs **Jordan chains**
- Correct object: **companion linearization**
- Appears as a **cluster**
 $\omega_{n,1}, \dots, \omega_{n,m}$
- Cluster size $\sim q_n^{1/p}$ ($p =$ max chain length)

simple $\implies$ one root; semisimple $\implies$ one subspace; defective $\implies$ one cluster

That is why the paper develops eigenvalue/eigenfunction theory (simple), eigenspace theory (semisimple), and cluster theory via companion linearization (defective).

What Do the Continuous Assumptions Buy Us?

We assume:

1. $\omega \mapsto \mathcal{P}(\omega)$ is holomorphic
2. Each $\mathcal{P}(\omega) : D \rightarrow X$ is Fredholm of index 0
3. There exists one regular point ω_* such that $\mathcal{P}(\omega_*)$ is invertible

Then analytic Fredholm theory gives:

- The resolvent set $\rho(\mathcal{P}) = \{\omega : \mathcal{P}(\omega) \text{ invertible}\}$ is open
- The resolvent map $\omega \mapsto \mathcal{P}(\omega)^{-1}$ is holomorphic on $\rho(\mathcal{P})$
- The spectrum $\sigma(\mathcal{P}) = \Lambda \setminus \rho(\mathcal{P})$ is **discrete** in the Fredholm region
- Every eigenvalue has **finite algebraic multiplicity**

Interpretation: This is why we are allowed to speak of isolated quasi-normal frequencies and local spectral components.

Before discretizing, we need to know that the continuous problem has a usable local spectral structure.

Why Contour Projectors?

Suppose $\Gamma \subset \rho(\mathcal{P})$ is a positively oriented contour enclosing one isolated spectral component.

In the **semisimple** case, define:

$$\Pi_\Gamma = \frac{1}{2\pi i} \int_\Gamma \mathcal{P}(\omega)^{-1} \mathcal{P}'(\omega) d\omega$$

What Π_Γ does:

- It is a **projection**
- Its range is the **local eigenspace**
- Its rank is the local **algebraic multiplicity**

Main philosophy:

Instead of comparing individual roots, compare the **continuous and discrete spectral projectors**.

This is more stable and better adapted to approximation theory.

projector convergence $\implies$ eigenspace convergence $\implies$ local spectral convergence
For spectral approximation, the contour projector is often the “right” object, more than the bare eigenvalue equation.

Chebyshev Collocation Discretization

Let $V_n = \Pi_{n-1} = \text{span}\{T_0, \dots, T_{n-1}\}$. Choose the Chebyshev–Gauss points:

$$y_j^{(n)} = \cos\left((2j-1)\frac{\pi}{2n}\right), \quad j = 1, \dots, n$$

Let $I_n : D \rightarrow V_n$ be the interpolation operator at these nodes.

Discrete collocation pencil:

$$\mathcal{P}_n(\omega) = I_n \mathcal{P}(\omega)|_{V_n} : V_n \rightarrow V_n$$

Discrete spectral problem:

Find $(\omega_n, u_n) \in \mathbb{C} \times (V_n \setminus \{0\})$ such that

$$\mathcal{P}_n(\omega_n)u_n = 0.$$

Matrix form:

If $u_n(y) = \sum_{k=0}^{n-1} c_k T_k(y)$, then

$$Q_n(\omega)c = 0$$

$$Q_n(\omega) = M_0^{(n)} + i\omega M_1^{(n)} + \omega^2 M_2^{(n)}$$

No boundary rows are added at the discrete level. Why? Because the boundary behaviour has already been built into the continuous admissible class D .

Why the Chebyshev–Sobolev Scale Appears

The right scale comes from the cosine change of variables: $y = \cos \theta$, $\theta \in [0, \pi]$.

If $U(\theta) = u(\cos \theta)$, then U is extended as an even 2π -periodic function.

Definition: The Chebyshev–Sobolev space $H_w^s(-1, 1)$ is defined by:

$$u \in H_w^s(-1, 1) \iff U \in H_{\text{even}}^s(\mathbb{T})$$

So H_w^s is the pullback of the ordinary periodic Sobolev space under $y = \cos \theta$.

Why this is useful: Chebyshev interpolation becomes trigonometric interpolation on a midpoint grid in θ . This makes it possible to use:

- Fourier coefficients and aliasing formulas
- Trigonometric interpolation estimates

Typical functional setting: $D = H_w^m(-1, 1)$, $X = H_w^r(-1, 1)$, $m > r + 2$.

The Chebyshev–Sobolev scale is not artificial. It is the **natural** Sobolev scale for Chebyshev approximation.

The Consistency Theorem

Consistency defect: $\varepsilon_n(K) = \sup_{\omega \in K} \|\mathcal{P}_n(\omega)I_n - I_n\mathcal{P}(\omega)\|_{D \rightarrow X}$, for compact $K \subset \Lambda$.

Algebraic consistency:

If $D = H_w^m$, $X = H_w^r$, $m > r + 2$, $r > 1/2$,
and $\mathcal{P}(\omega) : H_w^{r+2} \rightarrow H_w^r$ is uniformly
bounded on K , then:

$$\varepsilon_n(K) \leq C_K n^{r+2-m}$$

Analytic version:

If the eigenfunctions and coefficients are
analytic in a Chebyshev sense, then:

$$\varepsilon_n(K) \leq C_{K,\rho} n^{r+2} \rho^{-n}$$

Proof idea: pass to $y = \cos \theta \rightarrow$ Chebyshev interpolation becomes trigonometric
interpolation $\rightarrow$ use aliasing formula $\rightarrow$ estimate $I_n u - u$ in weaker norm $\rightarrow$ push error
through $\mathcal{P}(\omega)$.

regularity of the exact mode $\implies$ quantitative collocation consistency

From Consistency to Discrete Contour Stability

Fix a contour $\Gamma \subset \rho(\mathcal{P})$ around one isolated **semisimple** QNM ω_0 .

We already know: $M_\Gamma := \sup_{\omega \in \Gamma} \|\mathcal{P}(\omega)^{-1}\| < \infty$, $\varepsilon_n(\Gamma) = \sup_{\omega \in \Gamma} \|\mathcal{P}_n(\omega)I_n - I_n\mathcal{P}(\omega)\|$ small.

Can the discrete pencil also be invertible on the same contour for large n ?

Yes. If $\varepsilon_n(\Gamma)$ is sufficiently small, then $\Gamma \subset \rho(\mathcal{P}_n)$ for all large n , and $\sup_{\omega \in \Gamma} \|\mathcal{P}_n(\omega)^{-1}\| \leq C_\Gamma$.

Proof idea: Construct a discrete parametrix from the continuous inverse:

$$B_n(\omega) := I_n \mathcal{P}(\omega)^{-1} J_n$$

Then $\mathcal{P}_n(\omega)B_n(\omega) = I + \text{small error}$. If the error is $\{< 1\}$, invertibility follows by Neumann series.

consistency + continuous contour stability $\implies$ discrete contour stability

The Local Comparison Principle

Once both pencils are invertible on Γ , compare the discrete inverse lifted to the continuous space with the continuous inverse:

$$J_n \mathcal{P}_n(\omega)^{-1} I_n \quad \text{vs} \quad \mathcal{P}(\omega)^{-1}$$

Continuous projector:

$$\Pi = \frac{1}{2\pi i} \int_{\Gamma} \mathcal{P}(\omega)^{-1} \mathcal{P}'(\omega) d\omega$$

Lifted discrete projector:

$$\tilde{\Pi}_n = \frac{1}{2\pi i} \int_{\Gamma} J_n \mathcal{P}_n(\omega)^{-1} \mathcal{P}_n'(\omega) I_n d\omega$$

Then $\tilde{\Pi}_n - \Pi$ is controlled by:

- the resolvent-level difference
- the consistency defect
- the derivative defect

Proof idea: Subtract the two contour integrals, estimate the integrand uniformly on Γ .

The right local object to compare is the **contour projector**, not just the eigenvalue equation.

Projector Convergence

Under the Chebyshev–Sobolev assumptions, $\|\tilde{\Pi}_n - \Pi\| \rightarrow 0$.

Sobolev regularity H_w^m :

$$\|\tilde{\Pi}_n - \Pi\| \leq C_\Gamma n^{r+2-m}$$

Analytic regime:

$$\|\tilde{\Pi}_n - \Pi\| \leq C_{\Gamma,\rho} n^{r+2} \rho^{-n}$$

Why this is so important: if two projectors are close, their ranges are close. So projector convergence gives:

- Convergence of the local discrete **eigenspace**
- Preservation of local **algebraic multiplicity**
- And ultimately, local **spectral convergence**

projector convergence $\implies$ subspace convergence $\implies$ spectral convergence

This is the conceptual heart of the proof strategy.

The Main Convergence Theorem

Local spectral convergence near an isolated semisimple QNM.

Assume: ω_0 isolated semisimple eigenvalue of $\mathcal{P}$; $\Gamma \subset \rho(\mathcal{P})$ encloses ω_0 only; Chebyshev collocation assumptions hold.

Then, for all sufficiently large n :

1. $\Gamma \subset \rho(\mathcal{P}_n)$
2. The lifted discrete contour projector converges: $\tilde{\Pi}_n \rightarrow \Pi$
3. The enclosed algebraic multiplicity is preserved
4. All discrete eigenvalues of $\mathcal{P}_n$ inside Γ converge to ω_0

In the simple case: there is exactly one discrete eigenvalue ω_n inside Γ , and $\omega_n \rightarrow \omega_0$.

This is the first major theorem that justifies the numerical method: the discrete collocation spectrum near ω_0 approximates the continuous QNM.

This is where the whole chain of earlier results finally pays off.

First Consequences: Error Bounds and Validation

For a simple isolated QNM with $\mathcal{P}(\omega_0)u_0 = 0$, $\mathcal{P}(\omega_0)^*v_0 = 0$, and transversality $\tau_0 = (\mathcal{P}'(\omega_0)u_0, v_0)_X \neq 0$:

Eigenvalue estimate:

$$|\omega_n - \omega_0| \leq C\kappa(\omega_0)\eta_n$$

where

$$\kappa(\omega_0) = \frac{\|u_0\|_D \|v_0\|_X}{|\tau_0|}$$

is the **condition number** and η_n is the local collocation defect.

Towards certification: the theory can be turned into a **validation procedure** using continuous residuals, left and right approximate eigenfunctions, and contour-resolvent bounds.

A convergence theorem is not the end. It is the beginning of an error-analysis and validation theory.

Eigenspace estimates:

Projector convergence also gives:

- Eigenspace gap estimates
- Raw eigenfunction convergence
- Normalized eigenfunction convergence

Validated Spectral Windows

Big question: when we compute many discrete QNMs, which ones are trustworthy?

A single global theorem on the whole complex plane is too ambitious. Instead, we validate **window by window**.

Definitions on a compact set $K \subset \Lambda$:

Pollution-free: if $\omega_{n_j} \in \sigma(\mathcal{P}_{n_j}) \cap K$ and $\omega_{n_j} \rightarrow \omega_\infty \in K$, then $\omega_\infty \in \sigma(\mathcal{P})$.

Exhaustion: every isolated semisimple eigenvalue $\omega_* \in \sigma(\mathcal{P}) \cap K$ of multiplicity μ_* is approximated by exactly μ_* discrete eigenvalues for large n .

If both properties hold, K is a **validated spectral window**: no spurious accumulation points, every true isolated spectral component captured, multiplicities preserved.

How Validated Windows Work

Choose contours $\Gamma_1, \dots, \Gamma_M$ around the finitely many continuous eigenvalues in K , and let:

$$K_0 = K \setminus \bigcup_{j=1}^M \text{int}(\Gamma_j)$$

If: (i) $K_0 \subset \rho(\mathcal{P})$, (ii) the consistency defect $\rightarrow 0$ on K_0 , (iii) the local projector theorem holds on each Γ_j , then for all sufficiently large n :

1. $\sigma(\mathcal{P}_n) \cap K_0 = \emptyset$ (no spurious eigenvalues in the complement)
2. Each Γ_j contains the correct discrete multiplicity
3. Therefore K is a validated spectral window

Global validation is done on compact windows, not all at once.

What the Present Theory Gives

Local theory (strong):

Near an isolated semisimple QNM:

- Consistency of Chebyshev collocation
- Contour-resolvent stability
- Convergence of contour projectors
- Eigenvalue and eigenfunction error bounds
- Computable a posteriori validation

Defective theory (also strong):

Near a defective QNM:

- Companion-linearization framework
- Convergence of a discrete cluster
- Correct total algebraic multiplicity
- Fractional-power splitting laws
- Cluster recognition and subspace validation

The local theory is essentially complete: from problem formulation through error bounds and computable validation, for both semisimple and defective eigenvalues.

The Real Research Frontier

We do **not** have $\sigma(\mathcal{P}_n) \rightarrow \sigma(\mathcal{P})$ globally on the whole unbounded plane. **Why not?**

- **Finite-dimensional obstruction:** each $\mathcal{P}_n$ has finitely many eigenvalues; an infinite spectrum cannot be captured at fixed n
- **Lack of uniform large-window control:** would need uniform resolvent, consistency, and separation bounds as windows grow — generally unrealistic
- **Pseudospectral instability:** highly damped modes are most sensitive — exactly where uniform control is least plausible

The mathematically appropriate global statement is:

Spectral exactness on compact windows, and exhaustion by increasing windows.

This matches numerical practice: increase Chebyshev modes $\rightarrow$ validate a larger window $\rightarrow$ reveal more spectrum.

This is what real research looks like: the local theory may be complete and rigorous, while the global theory remains subtle, partly open, and deeply connected with the structure of the problem itself.

Backup Slides

Why Is Π_Γ a Projector? (semisimple case only)

Set $A_0 := \mathcal{P}(\omega_0)$, $A_1 := \mathcal{P}'(\omega_0)$. Near an isolated **semisimple** eigenvalue ω_0 , the resolvent has a **first-order pole**:

$$\mathcal{P}(\omega)^{-1} = \frac{R_{-1}}{\omega - \omega_0} + H(\omega), \quad H \text{ holomorphic.}$$

Hence $\mathcal{P}(\omega)^{-1} \mathcal{P}'(\omega) = \frac{R_{-1} A_1}{\omega - \omega_0} + \text{holomorphic}$, so the contour integral extracts the residue:

$$\Pi_\Gamma = \frac{1}{2\pi i} \int_\Gamma \mathcal{P}(\omega)^{-1} \mathcal{P}'(\omega) d\omega = R_{-1} A_1.$$

Resolvent identities: $A_0 R_{-1} = 0$, $R_{-1} A_0 = 0$, $A_0 R_0 + A_1 R_{-1} = I$

Idempotence:

$$\Pi_\Gamma^2 = R_{-1} A_1 R_{-1} A_1 = R_{-1} (I - A_0 R_0) A_1 = R_{-1} A_1 - \underbrace{R_{-1} A_0 R_0 A_1}_{=0} = \Pi_\Gamma.$$

Range = Eigenspace, Rank = Multiplicity (semisimple case only)

Range $\subset$ eigenspace:

$$A_0 \Pi_\Gamma = A_0 R_{-1} A_1 = 0 \implies \text{ran } \Pi_\Gamma \subset \ker \mathcal{P}(\omega_0).$$

Eigenspace $\subset$ range: if $u \in \ker A_0$, then from $R_0 A_0 + R_{-1} A_1 = I$:

$$u = (R_0 A_0 + R_{-1} A_1)u = R_{-1} A_1 u = \Pi_\Gamma u \in \text{ran } \Pi_\Gamma.$$

$$\text{ran } \Pi_\Gamma = \ker \mathcal{P}(\omega_0) \implies \text{rank } \Pi_\Gamma = \dim \ker \mathcal{P}(\omega_0)$$

In the **semisimple** case, algebraic multiplicity = geometric multiplicity = $\dim \ker \mathcal{P}(\omega_0)$, so:

$$\text{rank } \Pi_\Gamma = m_a(\omega_0).$$

Warning. For **defective** eigenvalues, the integral $(2\pi i)^{-1} \int_\Gamma \mathcal{P}(\omega)^{-1} \mathcal{P}'(\omega) d\omega$ is **not** a projection in general. The correct object is the contour projector of the **companion linearization**:

$$\mathbb{P} = \frac{1}{2\pi i} \int_\Gamma \mathbb{L}(\omega)^{-1} \mathbb{B} d\omega.$$

The Chebyshev–Sobolev Scale: Pullback Under $y = \cos \theta$

The key idea is simple: we are **not** inventing a strange Sobolev space on $(-1, 1)$. We take the **usual periodic Sobolev space** on the circle and transport it to the y -variable through $y = \cos \theta$.

Change of variables. Given $u = u(y)$ on $[-1, 1]$, define

$$U(\theta) := u(\cos \theta), \quad \theta \in [0, \pi].$$

Since $\cos(-\theta) = \cos \theta$ and $\cos(2\pi - \theta) = \cos \theta$, the function U extends to an **even 2π -periodic function** on $\mathbb{T}$.

Definition.

$$u \in H_w^s(-1, 1) \iff U = u(\cos \theta) \in H_{\text{even}}^s(\mathbb{T}).$$

$$H_w^s(-1, 1) = \mathcal{C}^{-1}(H_{\text{even}}^s(\mathbb{T})), \quad \text{where } \mathcal{C}u = u(\cos \theta)$$

Why the Pullback Is Natural

Chebyshev polynomials are cosines in disguise:

$$T_k(y) = \cos(k \arccos y) \implies T_k(\cos \theta) = \cos(k\theta).$$

So Chebyshev expansions in y become Fourier-cosine expansions in θ :

$$u(y) = \sum_{k \geq 0} \hat{u}_k T_k(y) \iff U(\theta) = \sum_{k \geq 0} \hat{u}_k \cos(k\theta).$$

The norm becomes transparent:

$$\|u\|_{H_w^s}^2 \asymp \sum_{k \geq 0} (1 + k^2)^s |\hat{u}_k|^2.$$

H_w^s is the space of functions whose Chebyshev coefficients have the same weighted ℓ^2 -decay as Fourier coefficients in H^s .

What the Chebyshev–Sobolev Norm Measures

Case $s = 0$: Since $d\theta = dy/\sqrt{1-y^2}$,

$$\|U\|_{L^2(\mathbb{T})}^2 \asymp \int_{-1}^1 |u(y)|^2 \frac{dy}{\sqrt{1-y^2}}.$$

So $H_w^0(-1, 1) = L^2\left((-1, 1), (1-y^2)^{-1/2} dy\right)$

What the Chebyshev–Sobolev Norm Measures

the classical **Chebyshev-weighted L^2 space**. That is why the subscript w .

Case $s = 1$: Differentiating $U(\theta) = u(\cos \theta)$ gives $U_\theta = -\sin \theta u'(y)$, so

$$\|U_\theta\|_{L^2(\mathbb{T})}^2 \asymp \int_{-1}^1 (1 - y^2)^{1/2} |u'(y)|^2 dy.$$

Case $s = 2$: Differentiating again: $U_{\theta\theta} = (1 - y^2)u''(y) - yu'(y)$.

The natural second-order operator $(1 - y^2)u'' - yu'$ transforms cleanly — a key reason these spaces are so well adapted to compactified QNM pencils.

Why Not Ordinary $H^s(-1, 1)$?

The cosine change of variables **degenerates at the endpoints** $y = \pm 1$.

In the y -variable, ordinary derivatives do not behave naturally as in the θ -variable. The Chebyshev–Sobolev scale captures exactly the regularity relevant for:

- Chebyshev expansions and interpolation
- Chebyshev collocation
- Transformed differential operators after $y = \cos \theta$

$H_w^s(-1, 1)$ is just the ordinary periodic Sobolev space in θ , rewritten in $y = \cos \theta$.
This is not an arbitrary choice. It is the correct space for the approximation problem.

The Consistency Defect: What It Measures

$$\varepsilon_n(K) = \sup_{\omega \in K} \|\mathcal{P}_n(\omega)I_n - I_n\mathcal{P}(\omega)\|_{D \rightarrow X}$$

This measures the **failure of the following diagram to commute**:

$$\begin{array}{ccc} & \mathcal{P}(\omega) & \\ D & \longrightarrow & X \\ \downarrow I_n & & \downarrow I_n \\ & \mathcal{P}_n(\omega) & \\ V_n & \longrightarrow & V_n \end{array}$$

- The **operator norm** $\|\cdot\|_{D \rightarrow X}$: worst-case relative error over all unit vectors $u \in D$
- The **supremum over $\omega \in K$** : uniform control on a compact set — essential when $K = \Gamma$ is a contour

Consistency means: “discretize-then-apply” and “apply-then-discretize” become closer as $n \rightarrow \infty$.

Why the Defect Reduces to Interpolation Error

Since $\mathcal{P}_n(\omega) = I_n \mathcal{P}(\omega) \mid_{V_n}$, we have $\mathcal{P}_n(\omega)I_n = I_n \mathcal{P}(\omega)I_n$. Therefore:

$$\mathcal{P}_n(\omega)I_n - I_n \mathcal{P}(\omega) = I_n \mathcal{P}(\omega)(I_n - I)$$

Applied to a function $u \in D$:

$$(\mathcal{P}_n(\omega)I_n - I_n \mathcal{P}(\omega))u = I_n \mathcal{P}(\omega)(I_n u - u)$$

The entire consistency defect is controlled by the **interpolation error** $I_n u - u$.

So the convergence theory reduces to:

1. Choose the right function spaces (H_w^s)
2. Prove interpolation estimates there (aliasing formula)
3. Push those estimates through $\mathcal{P}(\omega)$

That is why the **Chebyshev–Sobolev scale** appears.

What Consistency Means for an Exact Eigenpair

Suppose $\mathcal{P}(\omega_0)u_0 = 0$ (exact eigenpair). Then $I_n \mathcal{P}(\omega_0)u_0 = 0$, so:

$$\mathcal{P}_n(\omega_0)I_n u_0 = (\mathcal{P}_n(\omega_0)I_n - I_n \mathcal{P}(\omega_0))u_0$$

Hence:

$$\|\mathcal{P}_n(\omega_0)I_n u_0\|_X \leq \varepsilon_n(K) \|u_0\|_D \quad (\omega_0 \in K)$$

The interpolated exact eigenfunction $I_n u_0$ is an **approximate discrete eigenfunction**, with residual controlled by ε_n .

Algebraic regularity: $u_0 \in H_w^m$

$$\|\mathcal{P}_n(\omega_0)I_n u_0\|_X \leq C n^{r+2-m} \|u_0\|_{H_w^m}$$

Analytic regularity: $u_0 \in \mathcal{A}_\rho$

$$\|\mathcal{P}_n(\omega_0)I_n u_0\|_X \leq C n^{r+2} \rho^{-n} \|u_0\|_{\mathcal{A}_\rho}$$

Proof of Consistency: Steps 1–2

The “proof idea” on the main slide is a five-step chain. Here we unpack each step.

Step 1: Pass to $y = \cos \theta$.

Define $U(\theta) := u(\cos \theta)$, extended as an even 2π -periodic function. Then:

$$u \in H_w^s(-1, 1) \iff U \in H_{\text{even}}^s(\mathbb{T}), \quad \|u\|_{H_w^s} = \|U\|_{H^s(\mathbb{T})}.$$

After this change of variables, Chebyshev approximation becomes Fourier-cosine approximation, because $T_k(\cos \theta) = \cos(k\theta)$.

Step 2: Chebyshev interpolation becomes trigonometric interpolation.

The Chebyshev roots $y_j^{(n)} = \cos \theta_j^{(n)}$ with $\theta_j^{(n)} = (2j - 1)\pi/(2n)$ are the **midpoint grid** in θ . Under the cosine transform,

$$\mathcal{C}(I_n u) = \mathfrak{I}_n(\mathcal{C}u),$$

where $\mathfrak{I}_n$ is the trigonometric interpolation operator on the midpoint grid.

Polynomial interpolation in y is transported to trigonometric interpolation in θ — a classical, well-understood problem.

Proof of Consistency: Steps 3–4

Step 3: Use the aliasing formula.

The Fourier coefficients of the trigonometric interpolant satisfy:

$$(\widehat{\mathfrak{I}_n U})_k = \sum_{\ell \in \mathbb{Z}} (-1)^\ell \hat{U}_{k+2n\ell}, \quad |k| < n.$$

High frequencies **fold back** into the low-frequency band modulo $2n$, with alternating signs. This gives explicit control of the interpolation error in terms of the Fourier tail.

Step 4: Estimate $I_n u - u$ in a weaker norm.

We do **not** estimate $I_n u - u$ in the strong norm H_w^m . Instead, we work in H_w^{r+2} with the assumption $u \in H_w^m$, $m > r + 2$:

$$\|I_n u - u\|_{H_w^{r+2}} \leq C n^{r+2-m} \|u\|_{H_w^m}.$$

Stronger regularity of $u \implies$ more decay in Chebyshev/Fourier coefficients $\implies$ better interpolation error in a weaker norm.

Proof of Consistency: Step 5

Step 5: Push the error through $\mathcal{P}(\omega)$.

The key identity: $\mathcal{P}_n(\omega)I_n - I_n\mathcal{P}(\omega) = I_n\mathcal{P}(\omega)(I_n - I)$

Now chain the estimates for $u \in H_w^m$:

$$\begin{aligned}\|\mathcal{P}_n(\omega)I_n u - I_n\mathcal{P}(\omega)u\|_{H_w^r} &\leq \|I_n\|_{\mathcal{L}(H_w^r)} \|\mathcal{P}(\omega)(I_n u - u)\|_{H_w^r} \\ &\leq C_r M_K \|I_n u - u\|_{H_w^{r+2}} \\ &\leq C_K n^{r+2-m} \|u\|_{H_w^m}.\end{aligned}$$

interpolation stability $\times$ operator mapping bound $\times$ interpolation error = consistency rate

The hard part is purely approximation-theoretic: control $I_n u - u$. Once that is done on the right Sobolev scale, the pencil $\mathcal{P}(\omega)$ just transports the interpolation error into a consistency defect.

What Does “Analytic in the Chebyshev Sense” Mean?

Define $U(\theta) := u(\cos \theta)$, extended as an even 2π -periodic function. The **Chebyshev analytic class** $\mathcal{A}_\rho$ ($\rho > 1$) requires the Fourier coefficients to decay **geometrically**:

$$\|U\|_{\mathcal{A}_\rho} := \sup_{k \in \mathbb{Z}} \rho^{|k|} |\hat{U}_k| < \infty.$$

Since $T_k(\cos \theta) = \cos(k\theta)$, Fourier-cosine coefficients of U are Chebyshev coefficients of u . Hence:

analytic in the Chebyshev sense $\iff$ Chebyshev coefficients decay geometrically like ρ^{-k}

This is the right notion of analyticity for Chebyshev methods — it produces spectral (geometric) convergence.

From Chebyshev Analyticity to Spectral Convergence

If $u \in \mathcal{A}_\rho$, the analytic interpolation error theorem gives:

$$\|u - I_n u\|_{H_w^s} \leq C_{s,\rho} n^s \rho^{-n} \|u\|_{\mathcal{A}_\rho}$$

Pushing through $\mathcal{P}(\omega)$ as before yields the **spectral consistency rate**:

$$\varepsilon_n(K) \leq C_{K,r,\rho} n^{r+2} \rho^{-n}$$

For any $1 < \rho_0 < \rho$, the polynomial prefactor is absorbed into the geometric rate:

$$\varepsilon_n(K) \leq C \rho_0^{-n}$$

The logic is the same five-step chain as the algebraic case, but with geometric Fourier decay replacing algebraic Sobolev decay. Pass to $y = \cos \theta \rightarrow$ Chebyshev polynomials become cosines $\rightarrow$ “Chebyshev analyticity” = geometric coefficient decay $\rightarrow$ geometric convergence of interpolation and hence of the collocation defect.

The Discrete Parametrix: What and Why

A **parametrix** is an **approximate inverse**: an operator B such that $AB = I + E$ with E “small”.

The construction. We already know $\mathcal{P}(\omega)$ is invertible on Γ . Build from it an approximate inverse for the **discrete** operator:

$$B_n(\omega) := I_n \mathcal{P}(\omega)^{-1} J_n : V_n \rightarrow V_n$$
$$V_n \xrightarrow{J_n} X \xrightarrow{\mathcal{P}(\omega)^{-1}} D \xrightarrow{I_n} V_n$$

In words: (i) embed discrete data into the continuous space via J_n ; (ii) solve the continuous problem exactly; (iii) interpolate back to V_n .

$B_n(\omega)$ says: “solve the continuous problem first, then project back to the discrete space.”

Why the Parametrix Almost Inverts $\mathcal{P}_n(\omega)$

If discretization and application **commuted perfectly**, B_n would be exact:

$$\mathcal{P}_n(\omega)B_n(\omega) = \mathcal{P}_n(\omega) I_n \mathcal{P}(\omega)^{-1} J_n = I_n \mathcal{P}(\omega) \mathcal{P}(\omega)^{-1} J_n = I_n J_n = I_{V_n}$$

In reality the diagram does **not** commute. The error is:

$$\mathcal{P}_n(\omega)B_n(\omega) = I_{V_n} + E_n(\omega), \quad \text{where} \quad E_n(\omega) = (\mathcal{P}_n(\omega)I_n - I_n\mathcal{P}(\omega))\mathcal{P}(\omega)^{-1} J_n$$

Three factors control $\|E_n\|$:

ε_n	M_Γ	$\ J_n\ $
consistency defect	continuous resolvent bound	embedding constant
<i>(how far the diagram fails to commute)</i>	<i>(how much the inverse amplifies the defect)</i>	<i>(cost of lifting discrete data)</i>

From Approximate to Exact Inverse

If $\|E_n(\omega)\| \leq \varepsilon_n M_\Gamma \|J_n\| < 1$, then $I_{V_n} + E_n(\omega)$ is invertible by the **Neumann series**:

$$\left(I_{V_n} + E_n\right)^{-1} = \sum_{k=0}^{\infty} (-E_n)^k$$

Define the corrected inverse:

$$C_n(\omega) := B_n(\omega) \left(I_{V_n} + E_n(\omega)\right)^{-1}$$

Then:

$$\mathcal{P}_n(\omega) C_n(\omega) = \left(I_{V_n} + E_n\right) \left(I_{V_n} + E_n\right)^{-1} = I_{V_n}$$

Since V_n is finite-dimensional, a right inverse is automatically two-sided. Therefore $\mathcal{P}_n(\omega)$ is **invertible**.

continuous resolvent control $\xrightarrow{\text{parametrix}}$ approximate discrete inverse $\xrightarrow{\text{Neumann}}$ exact discrete inverse

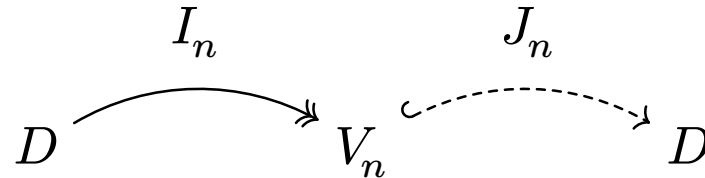
“Almost invertible” becomes “exactly invertible” after a Neumann correction.

The Operators I_n and J_n : Compression and Lifting

The continuous and discrete operators act on **different spaces**:

$$\mathcal{P}(\omega) : D \rightarrow X, \quad \mathcal{P}_n(\omega) : V_n \rightarrow V_n.$$

To compare them, we need two bridge operators going in **opposite directions**:



$$I_n : D \rightarrow V_n$$

interpolation / compression

“go from continuous to discrete”

$$J_n : V_n \hookrightarrow D$$

inclusion / lifting

“go from discrete to continuous”

Key property: $I_n J_n = I_{V_n}$ (interpolating something already in V_n changes nothing)

But: $J_n I_n \neq I_D$ ($J_n I_n$ is the interpolation projector onto V_n inside D)

Where J_n Appears in the Theory

1. Discrete coefficient operators:

$$\mathcal{L}_{j,n} := I_n \mathcal{L}_j J_n, \quad j = 0, 1, 2$$

Lift $v_n \in V_n$ to D via $J_n \rightarrow$ apply continuous operator $\mathcal{L}_j \rightarrow$ interpolate back via I_n .

2. Lifted discrete projector:

$$\tilde{\Pi}_n := J_n \Pi_n I_n$$

Compress to V_n via $I_n \rightarrow$ apply discrete projector $\Pi_n \rightarrow$ lift back to D via J_n .

Without J_n , the continuous projector Π and the discrete projector Π_n live on different spaces and **cannot be compared**.

3. Parametrix:

$$B_n(\omega) := I_n \mathcal{P}(\omega)^{-1} J_n$$

J_n makes the continuous inverse usable in the discrete argument.

The pair (I_n, J_n) is what allows the discrete world and the continuous world to talk to each other.